Enterprise architecture

2005 -> 2026

What changed in the digital enterprise - and what the 2005 paper already got right.

ONE ENTERPRISE, MANY CONNECTED DECISIONS

OUTCOMES
Why change?
Goals and measures

→

WORK
Capabilities
Processes and roles

→

DIGITAL
Data, apps
and services

→

PLATFORM
Cloud, devices
and operations

Measure outcomes; feed learning back into priorities and design.

THE SHORT ANSWER

The core idea did not expire: make relationships between business intent and information systems explicit. What changed is the operating landscape - cloud services, distributed software, continuous delivery, richer telemetry, broader risk governance, and AI-enabled capabilities.

TRACE THE LINKS | DESIGN FOR CHANGE | GOVERN THE WHOLE

Based on the supplied 2005 paper and public standards / technical references listed on page 6.

Educational synthesis; no individual author or university affiliation is included.

The 2005 model: connect the layers

The paper treats enterprise architecture as more than an IT inventory: it links business-related and information-systems artifacts, then makes those relationships modelable.

STRATEGY

Goals, products, performance measures

↓

ORGANIZATION

Structure, capabilities, responsibilities

↓

PROCESS + INFORMATION

Workflows, functions, information objects

↓

APPLICATIONS + TECHNOLOGY

Systems that support work and their technical base

THE METAMODEL JOB

Define the kinds of things being modeled and the links between them - including cross-layer dependencies.

THE IMPLEMENTATION JOB

Represent those constructs and links in a modeling platform so views, references, and reports can be created consistently.

A MESSAGE THAT STILL RESONATES

The paper explicitly calls for architecture to stay current and adaptable, provide stakeholder-oriented views, support an architecture-management process, and interface with operational development platforms. Those needs were already visible in 2005; today, the faster and more distributed environment makes them harder to ignore.

Source: supplied 2005 paper, especially its discussion of consistency, adaptability, views, and operational interfaces. [1]

From model repository to digital ecosystem

Selected milestones show why enterprise architecture now spans more operational change. Dates refer to the cited public source milestones, not to the start of every underlying technology.

2005 The paper links strategy, organization, applications, and other business / IS artifacts through a metamodel. [1]

2011 NIST publishes a cloud definition centered on network access, shared resources, rapid provisioning, and service models. [4]

2014 Kubernetes is open-sourced, reflecting the growing need to manage containerized, distributed workloads. [5]

2023 NIST releases the AI Risk Management Framework for managing risks across AI design, development, use, and evaluation. [8]

2024 NIST Cybersecurity Framework 2.0 adds a stronger enterprise governance emphasis; a generative-AI risk profile is also published. [7, 8]

2026 NIST posts a concept note for a trustworthy-AI profile focused on critical infrastructure - a concept note, not a finalized profile. [8]

THE DIRECTION OF TRAVEL

More external dependencies

More frequent change

More explicit risk ownership

A connected, changing system

A useful contemporary map is not just a stack of IT boxes. It shows value, people, work, information, external services, runtime, control, and feedback together.

PURPOSE + STAKEHOLDERS

Outcomes / customers / obligations / measures

BUSINESS CAPABILITIES + OPERATING MODEL

Processes / decisions / roles / partners

CHANNELS

Web, mobile, people, devices

APPLICATIONS

Products, apps, managed services

INTERFACES

APIs, events, data exchange

ECOSYSTEM

Suppliers, cloud, external platforms

DATA + INFORMATION

Meaning, quality, lineage, access

AI-ENABLED CAPABILITIES

Models, inputs, outputs, human oversight

CROSS-CUTTING CONTROLS

Security / privacy / identity / resilience / risk / compliance

PLATFORM + RUNTIME

Cloud / containers / networks / devices / data platforms

OBSERVE AND ADAPT

Traces + metrics + logs

Operational signals help teams understand system behavior. They can inform investment and design decisions - but telemetry alone does not prove business value.

Illustrative synthesis. Cloud, container, and telemetry terms follow sources [4]-[6]; control and AI-risk scope follows [7]-[8].

What changed - and what did not

Treat architecture as a decision aid: a disciplined way to understand consequences across connected parts of an enterprise.

STILL TRUE

Relationships matter
A system map is valuable when links between intent, work, information, and technology are explicit.

Consistency matters
Shared definitions help reveal conflicts, gaps, and duplication across views.

People need different views
A model should answer stakeholder questions, not just satisfy the modeling tool.

Architecture must evolve
A stale model is less useful; keeping it current was already a 2005 requirement.

EXPANDED BY 2026

The boundary is wider
Cloud providers, managed services, APIs, devices, and partners are part of the landscape.

Change is more continuous
Automated platforms can deploy and recover distributed workloads at scale.

Evidence can be operational
Traces, metrics, and logs provide signals about running systems.

Risk has broader scope
Cybersecurity governance and AI trustworthiness need enterprise-level attention.

A PRACTICAL FOUR-STEP READING

1. **START WITH THE OUTCOME**
   What needs to improve, and how will success be observed?

2. **TRACE THE DEPENDENCIES**
   Which roles, processes, data, applications, services, and suppliers are involved?

3. **MAKE CHANGE EXPLICIT**
   Show ownership, interfaces, assumptions, risks, and the intended evolution path.

4. **LEARN FROM OPERATION**
   Use evidence and feedback to revisit decisions; update models when reality changes.

A model is a purposeful simplification - not a complete copy of the enterprise and not a substitute for evidence.

Sources and reading notes

The timeline and comparisons combine the supplied 2005 paper with public framework and technical documentation available through November 2026.

[1] 2005 paper

A Comprehensive Enterprise Architecture Metamodel and Its Implementation Using a Metamodeling Platform (2005). Supplied PDF; used for the 2005 baseline and its stated future needs.

[2] TOGAF Standard, 10th Edition

The Open Group. Framework overview and guidance for configuring enterprise architecture practice.
https://www.opengroup.org/togaf

[3] ArchiMate modeling language

The Open Group. Overview of an open, independent language for describing and visualizing relationships across business domains.
https://www.opengroup.org/archimate-forum/archimate-overview

[4] NIST cloud definition

NIST SP 800-145 (2011). Definition and essential characteristics of cloud computing.
https://csrc.nist.gov/pubs/sp/800/145/final

[5] Kubernetes overview

Kubernetes documentation. Containerized workloads, declarative configuration, automation, scaling, and recovery.
https://kubernetes.io/docs/concepts/overview/

[6] OpenTelemetry

OpenTelemetry documentation. Vendor-neutral framework for generating, exporting, and collecting traces, metrics, and logs.
https://opentelemetry.io/docs/what-is-opentelemetry/

[7] NIST Cybersecurity Framework 2.0

NIST. Enterprise cybersecurity risk-management framework; published 2024.
https://www.nist.gov/cyberframework

[8] NIST AI Risk Management Framework

NIST. AI RMF 1.0 (2023), Generative AI Profile (2024), and 2026 concept note for a critical-infrastructure AI profile.
https://www.nist.gov/itl/ai-risk-management-framework

Scope note: this is an educational synthesis, not a claim that one framework or tool fits every organization.